Methods

Machine-learning analysis of blood metals and metabolic syndrome in NHANES 2007–2018

1. Study design, data source and reporting

This cross-sectional prediction-model development study uses six consecutive two-year cycles of the National Health and Nutrition Examination Survey (NHANES), 2007–2008 through 2017–2018. NHANES applies a complex, multistage probability design to sample the civilian non-institutionalized United States population, pairing a household interview with a standardized examination and laboratory assessment [1,2]. All data are public and were downloaded from the National Center for Health Statistics (https://wwwn.cdc.gov/nchs/nhanes/). The thirteen survey modules used are listed in Table S1; all were merged on the respondent sequence number (SEQN), with the demographic file defining the sampled-person universe. Reporting follows TRIPOD+AI [3] and STROBE [4]. The Mendelian randomization component is reported separately.

2. Study population and data cleaning

Figure 1 gives the data-cleaning sequence with the number excluded at each step and the number remaining. From 59,842 pooled participants we excluded 25,072 aged under 20 years, 372 pregnant participants (pregnancy alters waist circumference, blood pressure, lipids and glucose), 20,653 adults missing any measurement required to define the outcome (mainly because triglycerides and fasting glucose are restricted to the morning fasting subsample), 695 who did not meet the fasting criterion of 8 to 24 hours (identified by a zero fasting-subsample weight, which NCHS assigns to every such participant in every cycle), and 2,171 with no blood metal assayed, leaving 10,879 adults. Selenium and manganese were not assayed in 2007–2008 or 2009–2010; those participants were retained because they contribute complete lead, cadmium and mercury data, making the resulting absence structural rather than random (Section 5). Refusal and don't-know codes were recoded to missing using each item's own code width (for example 7 and 9 for one-digit items but 777 and 999 for the number of drinks), so that genuine answers of 7 or 9 drinks were kept.

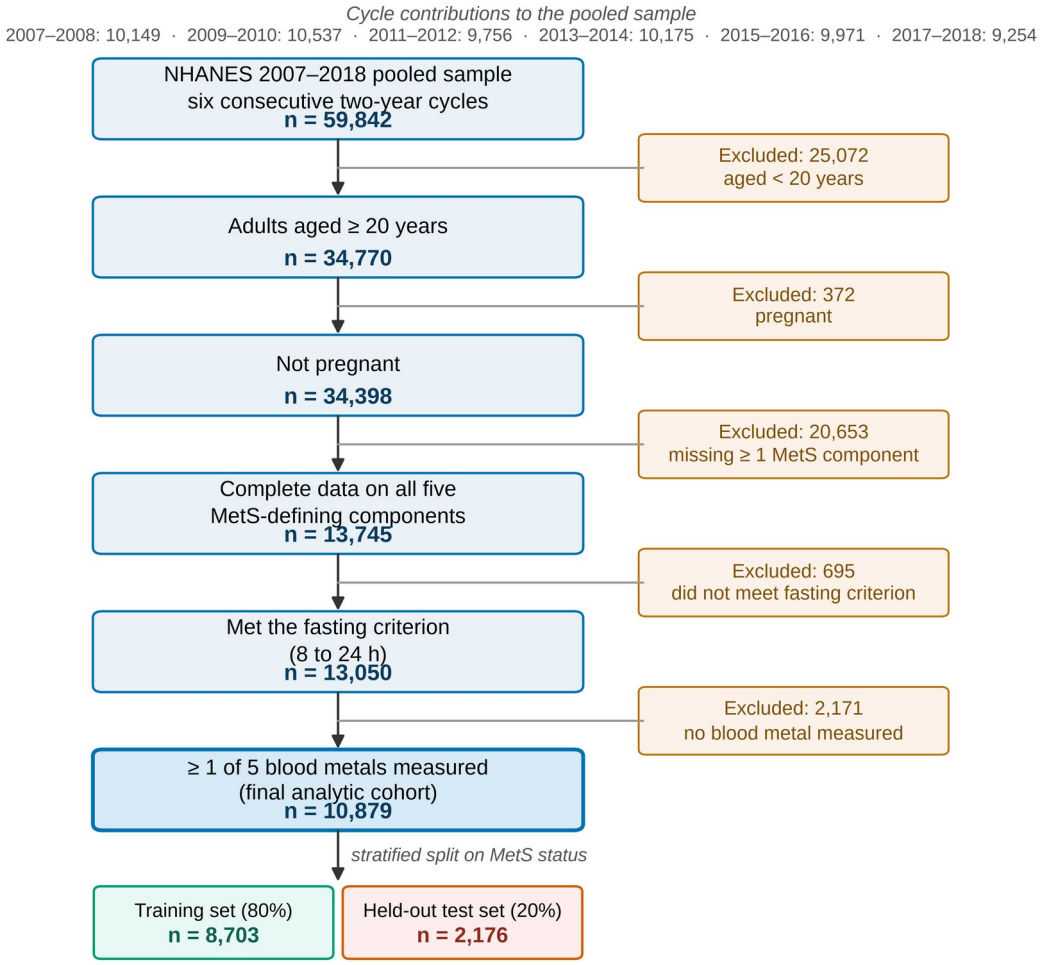


Figure 1. Data-cleaning and participant-flow diagram. Boxes on the left give the number remaining after each step; boxes on the right give the number excluded at that step and the reason. Cycle-level contributions to the pooled sample are shown above the diagram. MetS, metabolic syndrome.

3. Outcome definition

Metabolic syndrome was defined by the harmonized criteria of the joint interim statement of the International Diabetes Federation, NHLBI, American Heart Association and collaborating bodies [5], which retain the AHA/NHLBI thresholds and drug-treatment clauses [6]: a component was scored positive when the measured value met the cut-point or the participant reported drug treatment for it. Cases met at least three of five components; definitions, thresholds and source variables are in Table S2. Blood pressure was the mean of all valid readings (up to four; a diastolic reading of 0 mmHg was treated as invalid). NHANES asks about cholesterol-lowering treatment only once, without naming the drug, whereas the guideline's treatment clause for triglycerides and HDL cholesterol refers to lipid-specific drugs such as fibrates and nicotinic acid; we applied the item to both lipid components as a stated modelling choice and tested it in a

sensitivity analysis (Section 9). The treatment clauses together moved 1,237 participants from non-case to case and none in reverse, raising prevalence from 35.32% to 46.69%. The component measurements and medication items build the label only and were excluded from the predictor set, since a model given any of them could recover the outcome arithmetically rather than learn a risk relationship [7]. Body mass index is a predictor while waist circumference defines one component; the two are strongly correlated, so part of body mass index's predictive weight is close to definitional, and a model without it was fitted for comparison (Section 9).

4. Predictor variables

Predictors were specified in advance from the metabolic-syndrome and metal-toxicity literature, across four domains. The **blood-metal panel** is lead, cadmium, total mercury, selenium and manganese, the analytes measured together in the NHANES whole-blood assay and the panel used in prior NHANES analyses of metals and metabolic syndrome [8]. **Demographic and socioeconomic** predictors (age, sex, race/ethnicity, education, marital status, household size, income-to-poverty ratio, examination period) are established correlates of both the outcome and metal exposure and are retained as adjustment variables. The **anthropometric** domain is body mass index. **Lifestyle** predictors cover smoking, alcohol use, self-rated diet quality, four physical-activity items and resting pulse. Table S1 gives every variable with its NHANES item code, a plain description of what that code records, its coding and the model columns it contributes; after encoding the set held 32 columns from 24 source variables.

No data-driven selection procedure (stepwise regression or the least absolute shrinkage and selection operator) was used in this phase. Selection algorithms run on a fixed dataset produce model-selection instability, biased effect estimates and confidence intervals whose nominal coverage fails, so the surviving set partly reflects a stochastic screening step rather than which variables matter [9]. Since this phase exists to rank candidate exposures for the Mendelian randomization analysis, a step that discards variables before ranking would remove the quantity of interest. Penalized selection retains a clear role there, applied to the high-dimensional genetic instrument space rather than a pre-specified 32-column epidemiological set.

Two candidates were removed prospectively. Hip circumference was excluded on training-partition grounds: correlation 0.932 with body mass index and 81.3% missing, so it adds imputation burden while splitting importance with a near-identical retained variable. Physician-diagnosed diabetes was held out as partly downstream of the glucose component and

reintroduced only in sensitivity analysis. Three items needed harmonization or skip-pattern handling. The alcohol questionnaire was redesigned in 2017–2018, and its gatekeeper questions differ between periods ("at least 12 drinks in any one year" before, "ever had a drink" after), so alcohol use was represented by the construct closest to identical across cycles: any alcohol in the past 12 months, from the past-year drinking frequency, with average drinks per drinking day set to zero for past-year non-drinkers, who are never asked that question. The two versions route people differently around the frequency item (before 2017 those with fewer than 12 lifetime drinks skip it; from 2017 only lifelong abstainers do), so a few very light drinkers may be coded differently between periods. Current smoking intensity is asked only of people who have smoked at least 100 cigarettes, so never-smokers were coded as not smoking now. PAD645 (minutes walking or bicycling for transportation) is asked only of those answering yes to PAQ635, so a blank after a *no* denotes zero minutes and was recoded accordingly.

5. Missing data

After the skip patterns were resolved, missingness was concentrated in selenium and manganese (42.0% each, entirely the two cycles lacking the assay), the income-to-poverty ratio (9.1%) and the two alcohol items (5.9% and 5.9%); every other predictor was under 1%. Values were imputed inside the pipeline using the median for continuous and the modal category for categorical variables, fitted on training folds only. Multiple imputation is standard for etiological analysis [10] but sits awkwardly with a model that must also score a future incomplete record, and simulation work spanning development and deployment finds single imputation with missing indicators comparable for that purpose [11]. Each continuous predictor with missing values therefore carries a binary indicator (1 = missing and imputed); selenium and manganese share one indicator because they are missing for exactly the same participants, which makes it a marker of the 2007–2010 cycles rather than of individual non-response. The indicators are fixed in advance, so the transformed matrix holds a constant 37 columns in every fold. Indicators let the model use informative missingness but can bias predictions when missingness depends on the outcome [11]; they are reported separately from the substantive predictors rather than interpreted as risk factors.

6. Partitioning and prevention of information leakage

The cohort was split once into 8,703 training (80%) and 2,176 held-out test participants (20%), stratified on outcome. The test partition was opened only after development was complete and contributed to no imputation statistic, scaling parameter, resampling step, hyperparameter choice or decision threshold. With 4,063 events and 32 candidate columns the training partition provides about 127 events per candidate predictor; that benchmark was derived for regression models and does not translate directly to the effective complexity of a boosted ensemble [12], so it is a descriptive adequacy check rather than a licence for complexity, which was controlled by cross-validated tuning instead.

Every step that estimates anything from data sits inside one pipeline object, so cross-validation refits the whole sequence within each training fold (Figure S1). Fitting imputation, scaling or synthetic oversampling on the full dataset before splitting is among the most common forms of data leakage: a systematic survey identified 294 affected papers across 17 fields, and in several the corrected analysis erased the reported advantage of complex models over logistic regression [13]. Leakage through preprocessing and feature handling is specifically flagged as a recurring failure in biomedical machine learning [7]. The variable screens in Section 4 were likewise computed on training data alone.

7. Model development

Continuous predictors were median-imputed and standardized; binary, ordinal and one-hot columns were mode-imputed and left on their original scale. Standardization is required because the model set includes distance- and kernel-based learners (k-nearest neighbors, the support vector machine and the multilayer perceptron) whose objectives are not scale-invariant, so an unscaled predictor measured in hundreds (selenium, µg/L) would dominate the distance metric over one measured in units. Tree ensembles are invariant to monotone rescaling and are unaffected, so one shared preprocessing path keeps all eleven models comparable. Indicator columns are left unscaled because standardizing a 0/1 column changes nothing for trees and makes linear coefficients harder to read. Column order after transformation is deterministic and identical in every fold, which is what lets SHAP values be aligned unambiguously to predictor names.

Training-partition prevalence was 46.7%, so the classes are near-balanced and the usual case for resampling is weak. Synthetic Minority Over-sampling for Nominal and Continuous features

(SMOTENC) was nonetheless applied within each training fold. SMOTENC rather than SMOTE was necessary because the predictor set mixes continuous measurements with binary and one-hot columns: SMOTE interpolates linearly between neighbors and would place fractional values on categorical columns, creating synthetic records with impossible category memberships, whereas SMOTENC interpolates only continuous features and assigns each synthetic record the majority category among its neighbors [14]. The countervailing evidence is also relevant: imbalance correction can leave risk models badly miscalibrated without improving discrimination, and shifting the decision threshold achieves the same sensitivity–specificity trade-off without that cost [15]. We therefore treated the choice as a parameter to test rather than assume, compared SMOTENC against class weighting in a sensitivity analysis (Section 9), and report calibration alongside discrimination for every model.

Eleven classifiers spanning linear, instance-based, kernel, neural and tree-ensemble families were trained: logistic regression, decision tree, random forest, gradient boosting, LightGBM, XGBoost, CatBoost, support vector machine, Gaussian naïve Bayes, multilayer perceptron and k-nearest neighbors. Hyperparameters were tuned over the grids in Table S3 with 10-fold stratified cross-validation on the training partition, optimizing AUROC; every grid had at most 12 configurations and was searched exhaustively, and the selected configurations are reported there. Thresholds were selected by maximizing the Youden index [16] on out-of-fold training predictions, since choosing a threshold on the data used to report sensitivity and specificity inflates both; results are given at the tuned threshold and at 0.5.

8. Performance evaluation and model interpretation

Discrimination was summarized by AUROC and the area under the precision–recall curve, with accuracy, sensitivity, specificity, predictive values, F1 and Cohen's kappa at both thresholds; every test-set metric carries a 1,000-resample percentile bootstrap interval. Because all models are scored on the same participants their AUROC estimates are correlated, so comparing independent intervals would overstate the uncertainty of a difference; each model was compared with the leading model by DeLong's test for correlated AUROCs [17], with the interval of the difference from a paired bootstrap applying the same resampled indices to both models in every draw. Calibration was assessed with quantile-binned curves, the Brier score, the calibration slope, the calibration-in-the-large intercept and the observed-to-expected ratio, because a model can rank well while misstating absolute risk, the failure that matters if a

predicted probability reaches a clinician [18]. Clinical utility was assessed by decision-curve analysis against treat-all and treat-none strategies over threshold probabilities from 2% to 80% [19].

Predictor contributions were quantified two ways. SHapley Additive exPlanations were computed with the exact tree explainer [20], giving each predictor an additive contribution to each prediction on the log-odds scale, with global importance the mean absolute value. Permutation importance was the mean decrease in AUROC over ten permutations of each column, with the standard deviation retained so rank differences can be read against the noise producing them, following the permutation approach introduced for random forests and later formalized as model reliance [21,22]. The two measure different things, how far a predictor moves individual predictions, versus how much ranking performance depends on it, so a predictor substitutable by correlated predictors scores high on the first and low on the second, and both are reported. SHAP runs on the 37-column transformed matrix and permutation importance on the 32 original columns, so SHAP importances are ranked among those 32 predictors to keep one denominator. Cross-model agreement was assessed by recomputing permutation importance independently in each of the four leading models.

9. Sensitivity and subgroup analyses

Sensitivity analyses addressed the design choices most likely to affect conclusions: adding physician-diagnosed diabetes; replacing SMOTENC with class weighting (balanced sample weights for gradient boosting, which has no class-weight option); restricting the panel to blood lead alone; and three alternative outcome definitions (measured values only; no cholesterol-medication clause for the lipid components; blood pressure from the first reading only). Each used 10-fold cross-validation on the training partition with the three algorithms with the highest cross-validated AUROC (CatBoost, LightGBM and XGBoost) at their tuned hyperparameters, holding everything else fixed. The incremental value of the metals was assessed by refitting the leading model without the five metals, with lead as the only metal and without body mass index, and by a logistic regression on age and body mass index alone; each was scored on the same test participants and compared with the full model by DeLong's test. Because a single random partition leaves split-to-split variability uncaptured, all eleven models were also evaluated by repeated stratified 10-fold cross-validation (five repeats) on the training partition, reusing the tuned hyperparameters, so these estimates share the mild optimism of tuning on the same data.

Discrimination and calibration were also recomputed within strata of sex, age tertile and race/ethnicity, with bootstrap intervals and the calibration model refitted in every resample. Subgroup analyses are descriptive; each stratum is nested within the overall sample, so subgroup intervals are not independent of the overall estimate and are not used to test against it.

10. Scope, software and availability

This is a development study with internal validation: performance is estimated on a held-out partition of the same cohort, supplemented by repeated cross-validation and a temporal validation in which the leading algorithm was re-tuned and refitted on the 2007–2014 cycles and tested on the 2015–2018 cycles. No external cohort was available [3]. The models are a research instrument for prioritizing exposures for the Mendelian randomization phase, not a clinical screening tool, and have not been evaluated prospectively. The design is cross-sectional, so no association reported here can establish causation or temporal order; that question is left to the Mendelian randomization phase. Four further limitations follow from the design. Each metal was measured once in whole blood, which reflects recent exposure for lead and mercury and only partly reflects the long-term body burden of cadmium. The waist cut-points of 102 and 88 cm were applied to every participant, although the harmonized statement recommends lower population-specific values for people of Asian descent [5], whom the race/ethnicity variable used here does not separate. Body mass index and waist circumference overlap strongly, and the single cholesterol-medication item is applied to both lipid components, as described in Section 3. Analyses used Python 3.11–3.13 with scikit-learn, imbalanced-learn, LightGBM, XGBoost, CatBoost and SHAP, one seed fixed throughout. Descriptive statistics used the NHANES complex-survey design with Taylor-series linearization and domain estimation, the fasting-subsample weight divided by the number of pooled cycles (the analytic cohort is the fasting subsample), weighted geometric means for the metals, and design-based t and adjusted Wald F tests [1,2]; in 2013–2016 the metals were measured in a separate half-sample for which no combined weight exists, so metal estimates for those cycles are approximate; models were fitted unweighted, the target being individual-level discrimination rather than a population parameter. Data are public at https://wwwn.cdc.gov/nchs/nhanes/; code, derived metrics and model artifacts are kept under version control and will be deposited publicly on publication. NHANES protocols were approved by the NCHS Research Ethics Review Board with written informed consent; this secondary analysis required no additional approval.

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